

Notes: Gompers, Kaplan, and Mukharlyamov (JFE 2016)

What Do Private Equity Firms Say They Do?

Contents

1	Big picture: what the paper does and why it matters	3
2	Notation and “models” used in the paper	3
2.1	Core valuation identities	3
2.2	Discounted cash flow (DCF), WACC, and APV	4
2.3	IRR and MOIC (the dominant PE evaluation “model”)	4
2.4	Empirical classification models: k-means and factor analysis	5
3	Sample and survey design (Tables 1–4)	5
3.1	Table 1: PE firm respondents	5
3.2	Table 2: Enterprise value of portfolio companies	6
3.3	Table 3: Types of investments	6
3.4	Table 4: Individual respondents	6
4	Financial engineering I: deal evaluation and valuation (Table 5, Fig. 1, Tables 6–11)	7
4.1	Table 5: Deal evaluation metrics and methods	7
4.2	Figure 1: Forecast horizon	7
4.3	Table 6: Discount to management EBITDA forecasts	8
4.4	Table 7: Terminal value calculation and comparable selection	8
4.5	Table 8: IRR targets, determinants, and risk adjustments	9
4.6	Table 9: MOIC targets	10
4.7	Table 10: LP benchmark for performance	10
4.8	Table 11: Net IRR marketed to LPs	10
5	Financial engineering II: capital structure choices (Tables 12–13)	11
5.1	Table 12: Capital structure at closing	11
5.2	Table 13: What determines how much debt is used?	11

6	Governance engineering (Tables 14–18)	12
6.1	Table 14: Equity ownership and incentives	12
6.2	Table 15: Board size and composition	12
6.3	Table 16: Active involvement	13
6.4	Table 17: Recruiting management	13
6.5	Table 18: CEO replacement	14
7	Operational engineering and value creation (Tables 19–28)	14
7.1	Table 19: Deal funnel	14
7.2	Table 20: Deal sources and proprietary deals	14
7.3	Table 21: Deal selection factors	15
7.4	Table 22: Return drivers (importance and ranking)	15
7.5	Table 23: Pre-investment expected sources of value creation	17
7.6	Table 24: Who identifies pre-deal value?	17
7.7	Table 25: Post-investment sources of value creation	18
7.8	Table 26: Who identifies post-deal value?	18
7.9	Table 27: Types of exit	19
7.10	Table 28: Exit timing (importance and rank)	19
8	PE firm organization and operational resources (Tables 29–31)	19
8.1	Table 29: How PE firms organize internally	19
8.2	Table 30: Composition of investment professionals	19
8.3	Table 31: Use of external and internal advisers	20
9	Heterogeneity: firm types, clusters, factors, and founder backgrounds (Tables 32–34)	21
9.1	Table 32: Cluster analysis (k-means)	21
9.2	Table 33: Factor analysis	21
9.3	Table 34: Founder backgrounds and firm strategy	22
10	Concluding synthesis	24
11	Conclusion	24
11.1	Summary	24
11.2	Why PE practice differs from textbook finance	25
11.3	Takeaway	26

1 Big picture: what the paper does and why it matters

This paper uses a detailed survey of private equity (PE) investors to document *how PE firms claim to operate* along four dimensions:

- (i) valuation and deal evaluation metrics,
- (ii) capital structure choices (“financial engineering”),
- (iii) governance and management interventions (“governance engineering”),
- (iv) operational value creation (“operational engineering”).

The key contribution is *descriptive but economically sharp*: it directly measures practices that are hard to infer from deal outcomes alone. The survey evidence also allows the authors to document *heterogeneity in PE strategies* using cluster analysis and factor analysis, and then link those strategies to *founder backgrounds*.

Findings.

- PE investors overwhelmingly evaluate deals using **IRR** and **MOIC** (multiples), and rely heavily on **multiples-based terminal values** rather than CAPM-style DCF valuation.
- LP benchmarks are viewed as **mostly absolute** (net IRR or cash-on-cash), not risk-adjusted.
- Capital structure decisions incorporate both **trade-off** considerations and **market timing / debt market conditions**.
- Value creation is described as emphasizing **growth** and **operational improvements** more than “pure leverage engineering,” and governance interventions (including CEO replacement) are common.

2 Notation and “models” used in the paper

The paper is primarily survey-based, so it does not present a single structural model. However, it repeatedly refers to (and measures PE usage of) standard *valuation models* and *empirical classification models*. For study purposes, I treat these as the core “models” of the paper.

2.1 Core valuation identities

Enterprise value and equity value. Let

$$EV_t \equiv E_t + D_t - \text{Cash}_t,$$

where E_t is equity market value, D_t is net debt (gross debt net of cash), and Cash_t is cash (or treat D_t as net debt directly). In buyouts, leverage is commonly summarized by:

$$\text{Debt-to-capital} = \frac{D}{D + E}, \quad \text{Debt-to-EBITDA} = \frac{D}{\text{EBITDA}}.$$

EBITDA multiple valuation. If a comparable (market) multiple is M_t (e.g., $EV/EBITDA$), then a simple valuation is:

$$EV_t \approx M_t \cdot \text{EBITDA}_t.$$

In exit valuation, the terminal value frequently takes the form:

$$TV_T \equiv EV_T \approx M_T \cdot \text{EBITDA}_T.$$

2.2 Discounted cash flow (DCF), WACC, and APV

Even though the paper finds that most PE investors do not *use* DCF explicitly, the DCF objects are the standard academic benchmark.

WACC-based DCF. Let FCF_t be free cash flow to the firm. Then

$$EV_0^{DCF} = \sum_{t=1}^T \frac{\text{FCF}_t}{(1 + \text{WACC})^t} + \frac{TV_T}{(1 + \text{WACC})^T}.$$

A standard textbook WACC is

$$\text{WACC} = \frac{E}{D + E} R_e + \frac{D}{D + E} R_d (1 - \tau),$$

and CAPM equity cost is

$$R_e = R_f + \beta(\mathbb{E}[R_m] - R_f).$$

Adjusted present value (APV). The APV approach separates operating value (unlevered) from financing side effects:

$$EV_0^{APV} = V_0^U + PV(\text{tax shields}) - PV(\text{financing frictions}),$$

with

$$V_0^U = \sum_{t=1}^T \frac{\text{FCF}_t^U}{(1 + r_U)^t} + \frac{TV_T^U}{(1 + r_U)^T}.$$

2.3 IRR and MOIC (the dominant PE evaluation “model”)

Internal rate of return (IRR). Given net cash flows $\{CF_t\}_{t=0}^T$ (with $CF_0 < 0$ as the investment), IRR solves:

$$0 = \sum_{t=0}^T \frac{CF_t}{(1 + \text{IRR})^t}.$$

In practice, the paper documents that PE investors *target* an IRR hurdle (Table 8) and often apply ad hoc risk adjustments.

Multiple of invested capital (MOIC). MOIC is a level multiple:

$$\text{MOIC} = \frac{\sum_{t \geq 0} \max\{CF_t, 0\}}{-\sum_{t \geq 0} \min\{CF_t, 0\}}.$$

A simple connection (when cash flows are mainly initial outflow and terminal payoff) is:

$$\text{MOIC} \approx (1 + \text{IRR})^T,$$

but with interim distributions and reinvestment assumptions, the mapping is not one-to-one.

2.4 Empirical classification models: k-means and factor analysis

K-means (partition clustering). Given a vector of firm-level characteristics $x_i \in \mathbb{R}^p$ for firm i , k-means chooses clusters $C_1, \dots, C_K$ to minimize within-cluster variation:

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \mu_k\|^2, \quad \mu_k \equiv \frac{1}{|C_k|} \sum_{i \in C_k} x_i.$$

The paper reports a $K = 3$ solution (Table 32).

Factor analysis. A standard factor structure is:

$$x_i = \Lambda f_i + \varepsilon_i,$$

where f_i is a low-dimensional vector of latent factors, Λ is the loading matrix, and ε_i is idiosyncratic. The paper reports factor loadings for the first six factors (Table 33).

3 Sample and survey design (Tables 1–4)

3.1 Table 1: PE firm respondents

What is measured. Assets under management (AUM), recent fund performance (IRR and MOIC), firm age, and office location.

What I should remember.

- The survey covers **79 PE investors** with combined AUM exceeding **\$750bn** (paper abstract).
- Respondents span a wide AUM distribution (Table 1): the median AUM is far below the mean, indicating a right-skewed size distribution.
- The sample includes both **US-only** and **global** firms (Table 1), which is later used for subgroup comparisons throughout the tables.

Table 1

Private equity (PE) firm respondents.

This table describes the sample PE investor respondents. Reported are assets under management (AUM), performance of most recent fund (if available), and age of each private equity firm in the sample as of December 2012 from Preqin. Information on firms not covered by Preqin is taken from firm websites and the media. We also use the results of the current survey to determine office locations of firms in the sample. IRR=internal rate of return.

	N	Mean	25th percentile	Median	75th percentile	Standard deviation
AUM (millions of dollars)	79	9,548.6	750.0	3,400.0	11,000.0	15,021.1
IRR over benchmark (percent)	58	2.7	-3.9	0.9	6.9	11.8
Multiple of invested capital	58	1.3	1.2	1.3	1.5	0.3
Age (years)	79	19.5	12.0	19.0	26.0	10.5
Firms with						
- office(s) only in the US	44					
- office(s) outside the US	35					

Table 2

Enterprise value.

This table describes the enterprise value of portfolio companies of the sample private equity (PE) investors. Question is: "What fraction of the companies you invest in have the total enterprise value within the following ranges?" The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively.

Enterprise value	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
< \$25 million	11.8	0.0	19.0	5.5***	4.4	9.4	6.2	16.1**	16.2	6.3**
\$25 million – \$100 million	26.3	10.0	44.4	10.4***	16.8	23.5	20.1	31.0	33.4	17.5**
\$100 million – \$500 million	28.7	22.0	28.6	28.8	29.3	31.9	31.4	26.6	27.6	30.1
\$500 million – \$1 billion	16.8	10.0	5.2	27.0***	22.0	19.8	19.4	14.9	12.7	21.9**
> \$1 billion	16.4	0.0	2.8	28.3***	27.4	15.3	22.9	11.4**	10.2	24.1**
Number of responses	79	79	37	42	29	29	34	45	44	35

3.2 Table 2: Enterprise value of portfolio companies

What is measured. For each PE investor, the fraction of investments by enterprise value (EV) buckets.

Key patterns.

- Investments concentrate in **mid-market EV** buckets (Table 2): the largest fractions are in the \$25–100m and \$100–500m ranges.
- **Large-AUM firms** tilt strongly toward larger EV deals (Table 2), consistent with scale requirements and competition in mega-deals.
- Global vs local presence is also associated with systematically different EV exposure (Table 2), consistent with cross-border deal access and target size.

3.3 Table 3: Types of investments

What is measured. Whether the PE investor considers LBOs, growth equity, distress, PIPEs, etc. **Interpretation.**

- The “modal” PE investor in the sample is an **LBO investor** (Table 3 shows LBOs near-universal).
- Growth equity is also very common, while distress and PIPE activity is substantially more prevalent among larger or more global firms (Table 3).

3.4 Table 4: Individual respondents

What is measured. Titles/roles of the people filling out the survey. **Interpretation.**

- Responses come from **senior decision-makers** (general partners, managing partners, managing directors), supporting the view that answers reflect firm practices (Table 4).

Table 3

Type of investments.

This table describes the types of investments considered by the sample private equity (PE) investors. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. LBOs = leveraged buyouts; PIPEs = private investments in public equity.

Type of investment	Mean	AUM		IRR		Age		Offices	
		Low	High	Low	High	Old	Young	Local	Global
LBOs	90.1	82.9	97.2**	92.3	96.2	93.9	86.8	92.7	86.7
Growth equity	73.2	74.3	72.2	65.4	73.1	69.7	76.3	73.2	73.3
Distress	29.6	17.1	41.7**	26.9	34.6	30.3	28.9	19.5	43.3**
PIPEs	32.4	20.0	44.4**	38.5	38.5	45.5	21.1**	31.7	33.3
Other	8.5	8.6	8.3	3.8	7.7	9.1	7.9	7.3	10.0
Number of responses	71	35	36	26	26	33	38	41	30

Table 4

Private equity (PE) individual respondents.

This table describes the title of the individual filling out the survey at the sample PE investor firms. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively.

Title	Mean	AUM		IRR		Age		Offices	
		Low	High	Low	High	Old	Young	Local	Global
General partner or director	25.3	21.6	28.6	31.0	31.0	32.4	20.0	25.0	25.7
Managing partner	22.8	35.1	11.9**	13.8	20.7	20.6	24.4	29.5	14.3
Managing director	29.1	18.9	38.1	37.9	20.7	26.5	31.1	22.7	37.1
Chief financial officer	3.8	5.4	2.4	3.4	6.9	2.9	4.4	6.8	0.0
Other	19.0	18.9	19.0	13.8	20.7	17.6	20.0	15.9	22.9
Number of responses	79	37	42	29	29	34	45	44	35

4 Financial engineering I: deal evaluation and valuation (Table 5, Fig. 1, Tables 6–11)

4.1 Table 5: Deal evaluation metrics and methods

What is measured. Panel A: fraction of deals using each metric/method. Panel B: average ranking of reliance on each method.

Key patterns (Panel A).

- **IRR and MOIC are used in almost all deals** (Table 5).
- Comparable company multiples are also used frequently, while **APV/WACC DCF is rare** (Table 5).
- Larger firms appear more likely to use some additional metrics (e.g., free cash flow return to equity) (Table 5).

Key patterns (Panel B).

- **IRR is ranked as the dominant method**, far ahead of NPV/APV (Table 5).
- “Earnings multiple” approaches are the second most relied-upon class, consistent with multiples-based terminal values (Table 5).

Economic intuition. The results indicate that PE practice is organized around *hurdle-rate thinking* (IRR/MOIC) rather than discount-rate thinking (CAPM/WACC). Given the fund structure (finite horizon, illiquidity, and delegated LP–GP relationships), nominal hurdles can be a practical decision rule even if not fully consistent with standard asset pricing.

4.2 Figure 1: Forecast horizon

What is measured. Years of cash-flow forecasts used in PE models.

Interpretation.

Table 5

Deal evaluation metrics and methods.

Panel A presents the percentage of deals for which the sample private equity (PE) investors use different methods to evaluate an investment. Panel B provides the average ranking of different methods that the sample PE investors use to evaluate an investment, where 10 is the highest and 1 is the lowest. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. DCF=discounted cash flow; WACC=weighted average cost of capital; EBITDA=earnings before interest, taxes, depreciation, and amortization.

Metric/method	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Panel A: Deal evaluation metrics										
Gross IRR	92.7	100.0	88.5	97.0	99.9	96.9	100.0	86.4**	94.2	90.5
Multiple of invested capital	94.8	100.0	92.1	97.7	99.1	93.3	96.1	93.7	95.5	93.8
Adjusted present value (APV) DCF	9.3	0.0	7.1	11.5	9.3	7.3	3.9	13.9	10.5	7.6
WACC-based DCF	10.9	0.0	9.3	12.5	5.5	15.2	8.9	12.5	9.4	12.9
Comparable company EBITDA multiples	71.7	100.0	71.4	72.1	63.0	90.7**	75.9	68.1	76.7	64.8
Free cash flow return to equity	43.8	33.0	29.7	58.3***	45.2	43.7	44.4	43.2	40.7	48.1
Other	13.8	0.0	10.3	17.4	7.0	21.7	12.1	15.3	8.3	21.4
Number of responses	67	67	34	33	25	23	31	36	39	28
Panel B: Deal evaluation methods										
Accounting rate of return	0.6	0.0	0.9	0.2	0.7	0.7	0.5	0.7	0.6	0.5
Adjusted present value	0.9	0.0	0.6	1.2	0.6	0.8	0.5	1.3	0.8	1.0
Discounted payback period	1.7	0.0	1.7	1.6	1.3	1.3	1.3	1.9	1.9	1.3
Earnings multiple approach	6.1	8.0	6.0	6.1	5.5	6.2	5.4	6.7	6.1	6.1
Hurdle rate	3.6	0.0	3.7	3.5	4.4	2.0**	3.4	3.8	3.7	3.5
Internal rate of return	9.2	10.0	9.0	9.4	9.8	9.1	9.5	9.0	8.9	9.6
Net present value	2.8	0.0	2.1	3.4	3.7	2.0	2.5	3.0	2.4	3.3
Payback period	2.4	0.0	2.4	2.4	2.1	2.1	2.5	2.3	2.8	1.7
Profitability index	0.9	0.0	1.0	0.8	0.8	0.3	0.3	1.5**	0.5	1.4
Other	2.1	0.0	2.5	1.6	2.2	2.8	2.3	1.9	1.9	2.3
Number of responses	67	67	34	33	25	23	31	36	39	28

- Nearly all respondents use a **five-year forecast horizon** (Fig. 1).
- The terminal/exit value therefore carries substantial weight in valuation and is naturally implemented with exit multiples (link to Table 7).

4.3 Table 6: Discount to management EBITDA forecasts

What is measured. The haircut PE investors apply to management EBITDA forecasts.

Interpretation.

- PE investors apply a **large and systematic** discount (Table 6: mean roughly one-quarter; median roughly one-fifth).
- This highlights that the “value creation” margin is partly about **forecast discipline and information processing**, not only post-deal actions.

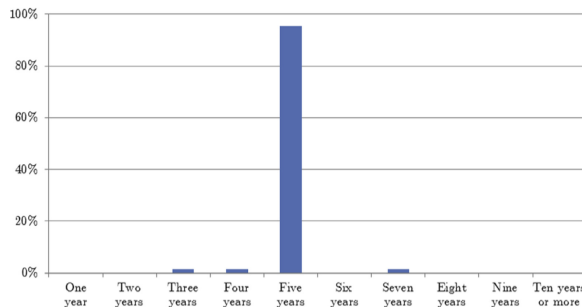


Fig. 1. Years of forecasts used by the sample private equity investors.

Table 6

Discount to management EBITDA forecasts.

This table reports the discount that the sample private equity (PE) investors normally apply to management EBITDA (earnings before interest, taxes, depreciation, and amortization) forecasts in their pro forma models. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence.

	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Discount to management EBITDA	0.25	0.20	0.27	0.21	0.26	0.25	0.21	0.28	0.25	0.25
Number of responses	44	44	27	17	15	14	20	24	29	15

4.4 Table 7: Terminal value calculation and comparable selection

Panel A: Terminal value methods.

Table 7

Terminal value calculation and comparable company selection.

Panel A reports the fraction of deals for which the sample private equity (PE) investors use different methods to calculate the exit value or terminal value of the model. Panel B shows the determinants of the selection that the sample PE investors use for comparable companies for multiples valuation or exit value. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. DCF=discounted cash flow.

	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Panel A: Terminal value calculation										
Comparable companies	81.4	100.0	75.3	87.7	81.3	88.6	87.3	76.3	78.2	85.8
Comparable transactions	71.4	99.0	67.8	75.0	73.2	80.3	79.5	64.4	76.1	64.8
DCF-based growing perpetuity	27.3	10.0	20.5	34.3	28.1	16.4	26.6	27.9	21.0	36.0
Other	25.6	0.0	33.5	17.4	20.7	31.1	22.8	27.9	28.7	21.3
Number of responses	67	67	34	33	25	23	31	36	39	28
Panel B: Comparable company selection										
Industry	96.9		96.9	97.0	95.8	100.0	100.0	94.1	94.6	100.0
Riskiness	49.2		34.4	63.6**	54.2	56.5	45.2	52.9	48.6	50.0
Size	84.6		87.5	81.8	87.5	87.0	90.3	79.4	86.5	82.1
Growth	73.8		75.0	72.7	79.2	69.6	77.4	70.6	75.7	71.4
Margins	66.2		59.4	72.7	83.3	65.2	67.7	64.7	62.2	71.4
Capital intensity	52.3		31.3	72.7***	62.5	52.2	51.6	52.9	51.4	53.6
Geography	56.9		43.8	69.7**	58.3	69.6	67.7	47.1	48.6	67.9
Other	4.6		6.3	3.0	4.2	8.7	3.2	5.9	8.1	0.0
Do not use comparables	3.1		3.1	3.0	4.2	0.0	0.0	5.9	5.4	0.0
Number of responses	65		32	33	24	23	31	34	37	28

- Terminal value is typically computed using **comparables** (public comps and transaction comps) (Table 7).
- A growing-perpetuity DCF terminal value is used by a minority (Table 7), reinforcing the limited role of DCF in practice.

Panel B: How comparables are selected.

- Industry and size are the most common comparability dimensions, followed by growth and margins (Table 7).
- Riskiness, capital intensity, and geography also matter for many respondents, especially among larger/global firms (Table 7).

4.5 Table 8: IRR targets, determinants, and risk adjustments

Panel A: Hurdle levels.

- The **median gross IRR target is about 25%** (Table 8).
- Smaller firms report **higher** targets than larger firms (Table 8), consistent with deal size, competition, and/or risk composition.

Panel B: Determinants of the target.

- Deal/firm **riskiness** is the dominant stated determinant (Table 8).
- Leverage and LP expectations also matter, but are secondary (Table 8).

Panel C: Risk adjustments.

- Many respondents adjust cash flows/IRR for **macro risks** (business cycle, interest rates, commodities, FX) and for **asset characteristics** (size, illiquidity) (Table 8).
- The pattern looks like an **ad hoc multi-factor risk model** rather than a pure CAPM/WACC approach.

Table 8

Internal rate of return (IRR) target, determinants, and adjustments.

Panel A reports the target value of gross IRR used by the sample private equity (PE) investors. Panel B describes the variables that the sample PE investors use to adjust their gross IRR target. Panel C reports the fraction of deals for which the sample PE investors adjust cash flows or the IRR to reflect different risks. The sample is divided into subgroups based on the median of assets under management (AUM), the IRR of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. LPs=limited partners; GDP=gross domestic product.

Variables	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Panel A: IRR										
Gross IRR target	27.0	25.0	30.0	24.1**	24.5	24.9	24.8	29.3	25.7	28.9
Number of responses	62	62	31	31	24	22	31	31	36	26
Panel B: IRR determinants										
Firm's riskiness	86.2		84.4	87.9	91.7	91.3	90.3	82.4	91.9	78.6
Leverage	47.7		40.6	54.5	58.3	52.2	51.6	44.1	54.1	39.3
Historical return expectations of LPs	30.8		40.6	21.2	20.8	30.4	22.6	38.2	37.8	21.4
Other	9.2		6.3	12.1	8.3	17.4	16.1	2.9	10.8	7.1
Not applicable	4.6		6.3	3.0	0.0	0.0	0.0	8.8	2.7	7.1
Number of responses	65		32	33	24	23	31	34	37	28
Panel C: Adjustments to the cash flows or the IRR										
Risk of unexpected inflation	17.7	0.0	8.2	26.9**	26.0	13.9	21.2	14.5	12.2	25.0
Interest rate risk	25.5	2.0	22.6	28.3	33.5	26.5	26.3	24.8	26.3	24.5
Term structure risk	18.5	0.0	16.6	20.3	14.9	26.9	22.9	14.4	13.5	25.0
GDP or business cycle risk	55.0	50.0	47.8	61.9	63.6	55.7	59.4	51.0	54.2	56.0
Commodity price risk	28.8	21.0	22.8	34.7	35.5	27.1	30.6	27.2	28.0	29.9
Foreign exchange risk	20.2	10.0	15.7	24.5	25.6	16.6	23.5	17.1	12.9	29.8***
Distress risk	13.0	0.0	8.7	17.2	13.8	11.9	17.2	9.1	9.0	18.2
Size	28.6	10.0	31.8	25.5	25.1	25.5	22.9	33.8	31.1	25.3
Market-to-book ratio	7.5	0.0	5.3	9.6	9.3	5.6	7.4	7.6	6.6	8.6
Momentum	11.8	0.0	9.6	13.9	17.0	10.3	18.9	5.4**	12.9	10.4
Illiquidity	20.3	0.0	22.2	18.5	19.8	6.8	15.2	25.0	15.8	26.3
Other	1.4	0.0	2.8	0.0	0.0	0.0	0.0	2.6	2.4	0.0
Number of responses	65	65	32	33	24	23	31	34	37	28

Table 9

Multiple of invested capital (MOIC).

This table measures the value of gross MOIC targeted by the sample private equity (PE) investors. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively.

	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Multiple of invested capital	2.85	2.50	3.16	2.54**	2.51	2.56	2.50	3.14**	2.98	2.67
Number of responses	62	62	31	31	24	21	28	34	36	26

4.6 Table 9: MOIC targets

What is measured. Target multiple of invested capital.

Interpretation.

- The median MOIC target is around **2.5x** (Table 9).
- Smaller and younger firms report higher MOIC targets (Table 9), consistent with strategy and competition differences.

4.7 Table 10: LP benchmark for performance

What is measured. The benchmark respondents believe LPs care about most.

Interpretation.

- Respondents report that LPs focus primarily on **absolute performance** (net IRR and cash-on-cash) (Table 10).
- Risk-adjusted public-market benchmarks (e.g., relative to the S&P 500) are rarely reported as “most important” (Table 10).

4.8 Table 11: Net IRR marketed to LPs

What is measured. Distribution of marketed net IRR ranges.

Interpretation.

Table 10

Benchmark for limited partners (LPs).

This table reports the most important benchmark for the LPs investing in the sample private equity (PE) investors. Net indicates net of all fees. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. S&P=Standard & Poor's; GPs=general partners.

Benchmark	Mean	AUM		IRR		Age		Offices	
		Low	High	Low	High	Old	Young	Local	Global
Net IRR	25.4	29.0	21.9	27.3	21.7	26.7	24.2	22.2	29.6
Net IRR versus S&P 500	7.9	6.5	9.4	9.1	4.3	10.0	6.1	8.3	7.4
Net IRR with respect to fund vintage year	27.0	19.4	34.4	27.3	43.5	40.0	15.2**	33.3	18.5
Net multiple or cash-on-cash	38.1	45.2	31.3	31.8	30.4	20.0	54.5***	33.3	44.4
IRR of other GPs	1.6	0.0	3.1	4.5	0.0	3.3	0.0	2.8	0.0
Number of responses	63	31	32	22	23	30	33	36	27

Table 11

Net internal rate of return (IRR) marketed to limited partners (LPs).

This table describes the typical IRR that the sample private equity (PE) investors market to their LPs. The sample is divided into subgroups based on the median of assets under management (AUM), the IRR of most recent fund, the age of PE investor, and by whether PE investor has a global presence.

Marketed IRR	Mean	AUM		IRR		Age		Offices	
		Low	High	Low	High	Old	Young	Local	Global
0–5%	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5–10%	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
10–15%	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
15–20%	15.6	18.8	12.5	13.0	13.0	23.3	8.8	18.9	11.1
20–25%	50.0	40.6	59.4	65.2	47.8	53.3	47.1	45.9	55.6
25–30%	23.4	25.0	21.9	21.7	30.4	20.0	26.5	24.3	22.2
> 30%	10.9	15.6	6.3	0.0	8.7	3.3	17.6	10.8	11.1
Number of responses	64	32	32	23	23	30	34	37	27

Table 12

Capital structure at closing.

This table reports the typical capital structure at closing for the sample private equity (PE) investor portfolio companies measured as debt-to-total capital and debt-to-EBITDA (earnings before interest, taxes, depreciation, and amortization). The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively.

Capital structure measure	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Debt-to-capital (percent)	55.7	60.0	54.3	57.2	56.6	56.9	55.0	56.4	55.0	56.8
Number of responses	62	62	31	31	22	23	30	32	37	25
Debt-to-EBITDA ratio	3.9	4.0	3.6	4.2**	4.1	4.2	4.2	3.6**	3.8	4.1
Number of responses	60	60	31	29	22	21	29	31	36	24

- The modal marketed net IRR is **20–25%** (Table 11).
- A meaningful share markets **25–30%** or **above 30%** (Table 11), consistent with high hurdle framing.

5 Financial engineering II: capital structure choices (Tables 12–13)

5.1 Table 12: Capital structure at closing

What is measured. Typical leverage at closing: debt-to-capital and debt-to-EBITDA.

Interpretation.

- Typical leverage is substantial (Table 12: debt-to-capital around 60% and debt-to-EBITDA around 4x).
- There is meaningful cross-sectional variation (Table 12), which motivates the later strategy heterogeneity analysis (Tables 32–34).

5.2 Table 13: What determines how much debt is used?

Panel A: Which factors are considered.

Table 13

Capital structure factors considered important and ranked.
Panel A reports the factors that the sample private equity (PE) investors consider in determining how much debt to raise for a transaction. Panel B reports the ranking of factors that the sample PE investors consider important in determining how much debt to raise for a transaction. A higher number means it is a more important factor with 6 being the highest rank. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively.

	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Panel A: Capital structure factors – important										
Ability of debt to force operational improvements	39.1		31.3	46.9	39.1	30.4	33.3	44.1	29.7	51.9
Use as much debt as the market will allow	65.6		65.6	65.6	65.2	69.6	73.3	58.8	62.2	70.4
Current interest rates and how much the company can pay	95.3		93.8	96.9	95.7	95.7	96.7	94.1	97.3	92.6
Industry that the firm operates in	96.9		93.8	100.0	95.7	95.7	93.3	100.0	97.3	96.3
Maximize trade-off between tax benefits and risk of default	67.2		59.4	75.0	65.2	78.3	66.7	67.6	64.9	70.4
Other	35.9		40.6	31.3	17.4	52.2**	36.7	35.3	43.2	25.9
Number of responses	64		32	32	23	23	30	34	37	27
Panel B: Capital structure factors – rank										
Ability of debt to force operational improvements	1.3	0.0	1.1	1.6	1.0	1.2	1.1	1.5	0.9	1.9**
Use as much debt as the market will allow	2.5	3.0	2.5	2.5	2.5	2.7	2.8	2.3	2.5	2.6
Current interest rates and how much the company can pay	5.0	5.0	4.7	5.3	5.1	4.9	5.0	4.9	4.9	5.0
Industry that the firm operates in	4.5	5.0	4.4	4.5	4.7	4.0	4.3	4.6	4.5	4.4
Maximize trade-off between tax benefits and risk of default	2.5	3.0	2.4	2.7	2.7	2.7	2.5	2.6	2.5	2.5
Other	1.8	0.0	2.1	1.4	0.8	2.7**	1.8	1.8	2.1	1.3
Number of responses	64	64	32	32	23	23	30	34	37	27

- **Debt market conditions** (interest rates and capacity to pay) and **industry** are almost universally important (Table 13).
- “Use as much debt as the market will allow” and “tax trade-off” considerations are important for many, but not at the top (Table 13).
- “Debt as discipline” (forcing operational improvements) is important for a minority (Table 13).

Panel B: Ranking of importance.

- The top-ranked factors are **market conditions** and **industry/fundamentals** (Table 13).
- This supports the paper’s interpretation: leverage choice reflects both **trade-off** and **market timing** logic.

6 Governance engineering (Tables 14–18)

6.1 Table 14: Equity ownership and incentives

What is measured. Typical equity ownership shares of PE investors, CEO, top management, etc.
Interpretation.

- PE investors typically retain **a controlling equity stake** (Table 14).
- CEOs and top managers receive meaningful equity stakes, consistent with strong incentive alignment.

6.2 Table 15: Board size and composition

What is measured. Desired board size distribution and average composition (insiders, PE directors, outside directors).

Interpretation.

- Boards are typically **small** (often 5–7 members) (Table 15).

Table 14

Typical equity ownership.

This table reports the typical equity ownership of the sample private equity (PE) investors, the chief executive officer (CEO), and top management. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively.

	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
PE investors	79.6	85.0	74.9	84.3**	82.7	83.6	82.9	76.6	81.2	77.3
CEO	8.0	5.0	10.0	6.0	7.1	6.1	7.8	8.2	6.9	9.5
Top ten management (excluding CEO)	7.2	7.0	8.1	6.3**	7.1	6.9	7.0	7.3	7.6	6.6
Other employees	1.8	0.0	1.1	2.4	3.0	0.9	1.7	1.8	1.3	2.5
Other	3.5	0.0	6.0	1.1**	0.1	2.6	0.6	6.1**	3.0	4.3
Number of responses	64	64	32	32	23	23	30	34	37	27

Table 15

Board of directors' size and composition.

Panel A presents the desired size of board of directors reported by the sample private equity (PE) investors. Panel B presents the desired composition of the board of directors by the sample PE investors. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively.

			AUM		IRR		Age		Offices	
	Mean	Median	Low	High	Low	High	Old	Young	Local	Global
Panel A: Board of directors' size										
3 or less	3.1		3.1	3.1	8.7	0.0	3.3	2.9	0.0	7.4
4	1.6		3.1	0.0	0.0	4.3	0.0	2.9	2.7	0.0
5	32.8		40.6	25.0	21.7	30.4	23.3	41.2	37.8	25.9
6	10.9		12.5	9.4	21.7	8.7	20.0	2.9**	10.8	11.1
7	46.9		37.5	56.3	39.1	52.2	46.7	47.1	43.2	51.9
8	3.1		0.0	6.3	8.7	0.0	3.3	2.9	2.7	3.7
9	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
10	1.6		3.1	0.0	0.0	4.3	3.3	0.0	2.7	0.0
11 or more	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Number of responses	64		32	32	23	23	30	34	37	27
Panel B: Board of directors' composition										
Inside directors	1.6	1.0	1.6	1.5	1.4	1.5	1.5	1.6	1.5	1.7
PE directors	2.8	3.0	2.7	2.9	2.8	2.9	2.8	2.7	2.7	2.8
Outside directors	1.7	2.0	1.6	1.9	1.9	1.7	1.9	1.6	1.8	1.6
Other	0.1	0.0	0.1	0.1	0.0	0.2	0.1	0.1	0.1	0.1
Number of responses	64	64	32	32	23	23	30	34	37	27

Table 16

Private equity (PE) involvement in portfolio companies.

This table reports the fraction of deals in which the sample PE investors become involved in the management of portfolio companies, i.e., actively advising the company on strategic choices. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence.

	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Active involvement										
Percent of deals	87.5	100.0	84.8	90.1	81.3	90.2	85.2	89.5	83.0	93.6
Number of responses	64	64	32	32	23	23	30	34	37	27

- PE directors represent a large fraction of seats, indicating strong control rights and active monitoring (Table 15).

6.3 Table 16: Active involvement

What is measured. Fraction of responses with active advisory/strategic involvement.

Interpretation.

- Active involvement is reported in **most deals**, with the median respondent indicating involvement in all deals (Table 16).

6.4 Table 17: Recruiting management

What is measured. Whether the firm recruits senior management before investing, after investing, or at either stage.

Interpretation.

- Most respondents do *not* recruit management *before* investing (Table 17), consistent with working with incumbent management to close deals.
- Roughly half recruit management *after* investing, and overall a majority recruit at some point (Table 17).
- Global firms are more likely to recruit after the deal (Table 17), consistent with larger organizational capacity and broader talent networks.

Table 17

Private equity (PE) recruitment of management teams.

This table reports the percentage of the sample PE investors who recruit their own senior management teams before investing, after investing, and before or after investing. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively.

Senior management recruitment	Mean	AUM		IRR		Age		Offices	
		Low	High	Low	High	Old	Young	Local	Global
Before investing									
Yes	31.3	31.3	31.3	26.1	39.1	30.0	32.4	29.7	33.3
No	68.8	68.8	68.8	73.9	60.9	70.0	67.6	70.3	66.7
After investing									
Yes	50.0	40.6	59.4	43.5	52.2	50.0	50.0	37.8	66.7**
No	50.0	59.4	40.6	56.5	47.8	50.0	50.0	62.2	33.3**
Before or after investing									
Yes	57.8	53.1	62.5	47.8	63.4	56.7	58.1	48.6	70.4*
No	42.2	46.9	37.5	52.2	36.6	43.3	41.9	51.4	29.6*
Number of responses	64	32	32	23	23	30	34	37	27

Table 18

Private equity (PE) replacement of chief executive officers (CEOs) after investing.

This table reports the percentage of deals in which the sample PE investors replace the CEO after the investment is made. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence.

CEO replaced	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Percent of deals	33.3	30.0	30.3	36.3	27.9	37.3	27.7	38.2	32.0	35.0
Number of responses	64	64	32	32	23	23	30	34	37	27

6.5 Table 18: CEO replacement

What is measured. Fraction of deals in which the CEO is replaced after investing.

Interpretation.

- CEO replacement is **common but not universal** (Table 18: roughly one-third of deals).
- This supports the view that governance engineering includes *both* incentive alignment and, when needed, managerial turnover.

7 Operational engineering and value creation (Tables 19–28)

7.1 Table 19: Deal funnel

What is measured. Transition rates from initial opportunities to due diligence, LOI, and closing.

Interpretation.

- The funnel is extremely selective: only a small fraction of considered opportunities close (Table 19).
- This emphasizes that value creation begins with **screening and selection**, not only post-deal actions.

7.2 Table 20: Deal sources and proprietary deals

What is measured. Shares of deals sourced through different channels, and the fraction of closed deals considered proprietary.

Interpretation.

- Deals come from a mix of self-generated sourcing and intermediated processes (Table 20).

Table 19

Deal funnel.

This table reports the percentage of opportunities considered by the sample private equity (PE) investors that reach different investment stages. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. LOI=letter of intent.

	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
All considered opportunities	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Deep due diligence	15.1	12.0	12.7	17.4*	17.6	13.7	16.5	13.9	14.5	15.9
Offer term sheet or negotiate detailed terms	12.9	10.0	11.4	14.4	14.6	12.1	13.2	12.6	12.5	3.4
Sign LOI	8.2	5.0	6.8	9.6	9.0	7.0	8.5	7.9	6.2	10.9**
Close	3.6	3.0	3.0	4.1*	4.1	2.9	3.7	3.5	3.0	4.3*
Number of responses	71	71	35	36	26	26	33	38	41	30

Table 20

Deal sources.

This table reports the percentage of the sample private equity (PE) investors who closed deals they identify from different sources and the percentage of closed deals the sample PE investors consider proprietary. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. LPs=limited partners.

Deal source	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Proactively self-generated	35.6	30.0	37.7	33.5	39.3	35.5	35.8	35.4	35.3	35.9
Investment bank-generated	33.3	25.0	30.5	36.0	33.0	37.0	38.2	29.1	33.0	33.7
Inbound from management	7.4	5.0	5.8	8.9	6.1	9.0	7.4	7.3	7.2	7.6
Other PE firm	4.3	0.0	5.4	3.3	4.1	2.2	4.1	4.5	5.0	3.5
Deal brokers	8.6	0.0	9.5	7.7	6.8	7.1	8.4	8.7	8.7	8.5
Executive network	8.6	5.0	8.3	9.0	8.8	8.5	4.6	12.1***	9.1	8.0
LPs or investors	1.7	0.0	2.5	0.9	0.6	0.7	0.4	2.7	1.2	2.3
Conferences	0.6	0.0	0.3	0.9	1.0	0.3	1.1	0.3	0.5	0.8
Other	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.1	0.1	0.0
Percent of closed deals considered proprietary	47.9	50.0	54.0	41.9**	48.0	43.9	41.5	53.4	47.6	48.3
Number of responses	71	71	35	36	26	26	33	38	41	30

- Respondents consider roughly **half of closed deals proprietary** (Table 20), though this is self-reported.
- Larger firms report less proprietary sourcing (Table 20), consistent with large targets being more likely to be auctioned.

7.3 Table 21: Deal selection factors

What is measured. Ranking of the importance of business quality, management, valuation, and ability to add value.

Interpretation.

- Business model/competitive position is ranked highest, followed by management and ability to add value (Table 21).
- Valuation is important, particularly among global firms (Table 21).

7.4 Table 22: Return drivers (importance and ranking)

What is measured. Which return drivers PE investors “bet on” and their ranked importance.

Interpretation.

- Growth and operational improvements are nearly universal and are ranked highest (Table 22).
- Leverage and multiple arbitrage matter for many, but are ranked as secondary (Table 22).
- The pattern aligns with a view of PE as **active owners** rather than pure financial engineers.

Table 21

Deal selection.

This table reports the ranking of factors considered by the sample private equity (PE) investors in choosing investments (where 6 is the highest rank). The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively.

Deal selection factor	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Ability to add value	3.6	3.0	3.6	3.6	3.9	3.5	3.5	3.7	3.5	3.8
Business model or competitive position	4.6	5.0	4.5	4.7	4.5	4.5	4.4	4.7	4.6	4.5
Fit with fund	2.3	2.0	2.7	1.9	2.0	2.2	2.2	2.4	2.5	2.0
Industry or market	3.2	3.0	3.4	3.0	3.3	3.0	3.3	3.1	3.5	2.8
Management team	3.8	4.0	3.7	4.0	3.9	4.0	4.1	3.6	3.8	3.8
Valuation	3.5	3.0	3.2	3.8	3.4	3.8	3.5	3.5	3.0	4.1***
Number of responses	65	65	32	33	24	23	31	34	37	28

Table 22

Return driver importance and ranking.

Panel A reports the percentage of the sample private equity (PE) investors who mention different return drivers that they bet on in making investments. Panel B reports the return drivers that the sample PE investors bet on in making investments ranked in order of importance where 6 is the highest rank. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively.

	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
<i>Panel A: Return driver – important</i>										
Growth in the value of the underlying business	100.0		100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Industry-level multiple arbitrage	64.8		74.3	55.6	50.0	65.4	66.7	63.2	61.0	70.0
Leverage	76.1		68.6	83.3	65.4	88.5**	81.8	71.1	73.2	80.0
Operational improvements	97.2		94.3	100.0	100.0	96.2	100.0	94.7	95.1	100.0
Refinancing	36.6		28.6	44.4	34.6	42.3	45.5	28.9	29.3	46.7
Other	26.8		28.6	25.0	23.1	19.2	21.2	31.6	31.7	20.0
Number of responses	71		35	36	26	26	33	38	41	30
<i>Panel B: Return driver – rank</i>										
Growth in the value of the underlying business	5.7	6.0	5.8	5.7	5.9	5.5**	5.7	5.7	5.8	5.6
Industry-level multiple arbitrage	2.4	3.0	2.8	2.1	1.9	2.4	2.3	2.6	2.1	2.9
Leverage	2.6	3.0	2.4	2.8	2.3	3.2	3.0	2.2**	2.5	2.7
Operational improvements	4.6	5.0	4.5	4.7	4.9	4.6	4.7	4.5	4.5	4.7
Refinancing	1.0	0.0	0.8	1.3	0.9	1.3	1.2	0.9	0.9	1.2
Other	1.1	0.0	1.2	1.0	0.7	0.8	0.8	1.3	1.3	0.8
Number of responses	71	71	35	36	26	26	33	38	41	30

Table 23

Pre-investment (expected) sources of value creation.

This table describes the percentage of deals that the sample private equity (PE) investors identify having specified pre-deal sources of value. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. IT=information technology; CEO=chief executive officer; CFO=chief financial officer.

Sources of value	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Reduce costs in general	35.6	27.5	35.8	35.5	37.1	37.3	39.9	32.0	31.0	41.8
Improve IT or information systems	26.1	20.0	30.8	21.6	22.0	23.3	23.9	28.0	26.7	25.3
Introduce shared services	15.6	2.5	16.4	14.9	11.6	18.3	16.9	14.6	14.9	16.6
Increase revenue or improve demand factors	70.3	80.0	77.5	63.5**	75.0	63.5	67.0	73.2	70.6	70.0
Redefine the current business model or strategy	33.8	29.5	27.8	39.5	43.0	29.8	32.1	35.3	32.8	35.2
Change CEO or CFO	30.6	27.5	33.4	28.0	29.2	32.9	30.9	30.4	29.3	32.4
Change senior management team other than CEO and CFO	33.4	30.0	37.3	29.7	32.5	33.1	27.9	38.1	35.4	30.8
Improve corporate governance	47.0	37.0	52.4	41.9	40.1	45.5	39.4	53.5	47.3	46.6
Improve incentives	61.1	73.5	60.7	61.5	58.3	67.0	65.5	57.4	59.0	63.9
Follow-on acquisitions	51.1	50.0	53.9	48.4	52.0	46.9	51.0	51.2	53.2	48.3
Strategic investor	15.6	10.0	16.4	14.8	12.3	14.0	14.4	16.5	15.1	16.2
Facilitate a high-value exit	50.0	43.5	61.0	39.6**	45.6	42.0	40.4	58.1**	53.5	45.4
Purchase at an attractive price (buy low)	44.3	43.0	49.2	39.6	38.2	43.3	40.9	47.1	44.9	43.5
Purchase at an attractive price relative to the industry	46.6	50.0	54.5	39.2**	38.7	47.3	42.9	49.8	50.1	42.0
Other	9.8	0.0	9.4	10.2	0.0	14.3**	9.4	10.1	12.4	6.4
Number of responses	74	74	36	38	27	27	34	40	42	32

Table 24

Pre-investment value creators.

This table reports the percentage of deals that each specified group actively participates in identifying pre-deal value for the sample private equity (PE) investors. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively.

Participants	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Deal team	97.7	100.0	96.9	98.4	97.7	99.2	98.4	97.0	97.9	97.4
Operating partners	45.3	40.5	44.9	45.7	46.9	46.3	40.5	49.4	41.6	50.2
Outside consultants	36.8	26.5	27.9	45.1**	35.0	45.3	42.1	32.2	35.0	39.0
Other	7.2	0.0	8.9	5.5	5.1	5.2	4.1	9.8	8.8	5.0
Number of responses	74	74	36	38	27	27	34	40	42	32

7.5 Table 23: Pre-investment expected sources of value creation

What is measured. Fraction of deals with each pre-deal value creation source identified.

Interpretation.

- Expected sources emphasize **revenue growth**, **incentives**, **governance**, and **follow-on acquisitions** (Table 23).
- “Buy low / facilitate a high-value exit” channels are also frequently mentioned (Table 23), mixing selection and timing narratives.

7.6 Table 24: Who identifies pre-deal value?

What is measured. Which participants identify pre-deal value opportunities.

Interpretation.

- Deal teams are involved almost always, while operating partners and consultants are involved in large minorities of deals (Table 24).
- Larger firms use consultants more, consistent with scale and ability to pay for external expertise.

Table 25

Post-investment sources of value creation.

This table reports the percentage of deals that the sample private equity investors identify as having specified post-deal sources of value and the difference from pre-deal sources of value from Table 23. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. IT=information technology; CEO=chief executive officer; CFO=chief financial officer.

Sources of value	Mean	Median	Δ from pre-deal	AUM		IRR		Age		Offices	
				Low	High	Low	High	Old	Young	Local	Global
Reduce costs in general	47.4	48.5	11.7	46.1	48.5	46.5	51.2	52.1	43.3	40.2	56.8**
Improve IT or information systems	33.5	28.0	7.4	36.4	30.6	29.7	35.3	31.0	35.5	32.8	34.3
Introduce shares services	21.9	10.0	6.3	18.5	25.2	21.9	24.0	23.9	20.3	18.8	26.1
Increase revenue or improve demand factors	69.5	71.0	-0.8	73.9	65.3	70.7	68.8	67.2	71.4	69.4	69.6
Redefine the current business model or strategy	40.1	40.0	6.3	34.2	45.7**	52.1	35.0**	39.3	40.8	39.7	40.7
Change CEO or CFO	42.9	40.0	12.3	40.5	45.3	46.3	43.8	44.1	42.0	40.6	46.0
Change senior management team other than CEO and CFO	47.1	50.0	13.7	46.2	48.0	44.1	52.6	46.7	47.4	48.2	45.7
Improve corporate governance	52.1	50.0	5.1	56.2	48.2	51.0	52.1	49.9	54.0	53.6	50.1
Improve incentives	65.1	71.5	3.9	58.3	71.5	70.3	72.3	72.3	59.0	60.5	71.1
Make follow-on acquisitions	48.1	50.0	-3.0	45.1	50.8	47.8	46.7	50.4	46.1	50.7	44.6
Bring on a strategic investor	13.5	10.0	2.1	14.5	12.5	14.1	10.1	13.3	13.6	15.4	10.9
Facilitate a high-value exit	58.8	60.0	8.8	62.7	55.0	55.6	53.6	55.9	61.2	62.8	53.5
Other	7.1	0.0	7.1	8.3	5.9	0.0	7.1	5.6	8.3	9.1	4.3
Number of responses	74	74	74	36	38	27	27	34	40	42	32

Table 26

Post-investment value creators.

This table reports the percentage of deals that each specified group actively participates in identifying post-deal value. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of private equity (PE) investor, and by whether PE investor has a global presence.

Participants	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Deal team	93.3	100.0	90.6	95.9	93.7	95.6	93.5	93.2	92.1	94.9
Operating partners	51.1	50.0	46.8	55.1	56.2	48.5	45.7	55.6	43.6	60.9
Outside consultants	27.1	21.0	22.0	31.9	26.2	34.0	29.3	25.2	25.2	29.5
Other	8.6	0.0	13.0	4.3	3.7	7.9	5.6	11.1	9.2	7.7
Number of responses	74	74	36	38	27	27	34	40	42	32

Table 27

Types of exit.

This table reports the fraction of deals the sample private equity (PE) investors target for different types of exit. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. IPO=initial public offering.

Type of exit	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
IPO	18.8	11.7	10.9	26.4***	23.7	18.9	20.6	17.2	12.1	27.7***
Strategic sale	51.0	50.0	57.3	44.8**	46.3	51.4	44.2	56.7**	57.5	42.3***
Financial sale	29.5	30.0	31.8	27.3	29.6	28.1	33.6	26.0	30.4	28.3
Other	0.7	0.0	0.0	1.5	0.5	1.6	1.6	0.0	0.0	1.7
Number of responses	63	63	31	32	22	23	29	34	36	27

7.7 Table 25: Post-investment sources of value creation

What is measured. Fraction of deals with each post-deal value source, and the difference relative to pre-deal expectations.

Interpretation.

- Several governance/operational actions are **more prevalent post-deal than pre-deal**, including cost reductions and management changes (Table 25).
- Revenue growth remains central, but the post-deal shift highlights that PE discovers and implements additional operational changes after control is obtained.

7.8 Table 26: Who identifies post-deal value?

What is measured. Post-deal value identification participants.

Interpretation.

- Operating partners are involved in **about half** of deals post-investment (Table 26), reinforcing operational engineering capacity.
- Consultant use is lower post-deal than pre-deal (compare Table 26 to Table 24), consistent with internal execution resources.

7.9 Table 27: Types of exit

What is measured. Targeted shares of exits via IPO, strategic sale, financial sale, etc.

Interpretation.

- Strategic sales are the most common targeted exit route, followed by financial sales; IPOs are less common but non-trivial (Table 27).
- Larger and global firms target IPO exits more frequently (Table 27), consistent with larger deals and global capital market access.

7.10 Table 28: Exit timing (importance and rank)

What is measured. Factors used to time exits and their importance ranks.

Interpretation.

- Exit timing balances **fundamentals** (achieving the operational plan) and **market timing** (capital market conditions), which are both top-ranked (Table 28).
- Hitting an IRR/ROI target and management’s opinion are also important, consistent with delegated incentives and relationship management.

8 PE firm organization and operational resources (Tables 29–31)

8.1 Table 29: How PE firms organize internally

What is measured. Whether the firm is organized by industry, product, criteria, as generalists, etc.

Interpretation.

- Industry specialization is very common; pure generalists are a minority (Table 29).
- Larger/global firms are more likely to be organized by product (Table 29), consistent with multi-product platforms.

8.2 Table 30: Composition of investment professionals

What is measured. Percentages of staff devoted to deal work, sourcing, execution, operating, shared services, fundraising, capital markets, etc.

Interpretation.

Table 28

Exit timing, importance and rank.

Panel A describes the factors the sample private equity (PE) investors consider in deciding on the timing of exit. Panel B describes the ranking of factors the sample PE investors consider in deciding on the timing of exit (where higher rank is more important and 7 is the highest rank). The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. ROI=return on investment; LPs=limited partners.

Factors	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Panel A: Exit timing – important										
Achieve operational plan set out to achieve	92.2		90.6	93.8	95.7	91.3	90.0	94.1	91.9	92.6
Capital market conditions	96.9		96.9	96.9	100.0	95.7	100.0	94.1	94.6	100.0
Competitive considerations	76.6		75.0	78.1	69.6	78.3	80.0	73.5	83.8	66.7
Hit IRR or ROI target	79.7		71.9	87.5	91.3	82.6	76.7	82.4	75.7	85.2
LPs pressure to return capital	56.3		46.9	65.6	56.5	56.5	43.3	67.6	48.6	66.7
Management's opinion	85.9		81.3	90.6	87.0	82.6	86.7	85.3	86.5	85.2
Other	14.1		15.6	12.5	13.0	13.0	13.3	14.7	10.8	18.5
Number of responses	64		32	32	23	23	30	34	37	27
Panel B: Exit timing – rank										
Achieve operational plan set out to achieve	5.5	6.0	5.4	5.5	5.8	5.3	5.3	5.6	5.5	5.4
Capital market conditions	5.3	5.5	5.4	5.2	5.3	5.3	5.4	5.2	5.0	5.7
Competitive considerations	3.5	4.0	3.5	3.5	3.2	3.1	3.4	3.5	3.9	2.9
Hit IRR or ROI target	4.0	4.0	3.9	4.2	4.2	4.6	3.8	4.3	3.9	4.3
LPs pressure to return capital	1.8	1.5	1.5	2.0	1.9	1.8	1.4	2.1	1.6	1.9
Management's opinion	3.7	4.0	3.4	4.1	3.8	3.7	4.3	3.2**	3.8	3.7
Other	0.7	0.0	0.7	0.7	0.7	0.7	0.6	0.7	0.5	0.9
Number of responses	64	64	32	32	23	23	30	34	37	27

Table 29

Private equity (PE) firm organization.

This table describes how the sample PE investors say their firm is organized. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively.

Firm organization	Mean	AUM		IRR		Age		Offices	
		Low	High	Low	High	Old	Young	Local	Global
Industry	54.4	54.1	54.8	65.5	55.2	64.7	46.7	54.5	54.3
Criteria	11.4	10.8	11.9	13.8	3.4	11.8	11.1	6.8	17.1
Product	16.5	5.4	26.2**	17.2	17.2	29.4	6.7***	2.3	34.3***
Generalist	36.7	40.5	33.3	27.6	34.5	26.5	44.4	43.2	28.6
Other	6.3	2.7	9.5	10.3	6.9	8.8	4.4	2.3	11.4
Number of responses	79	37	42	29	29	34	45	44	35

- The majority of staff are deal-oriented, but a meaningful fraction is dedicated to **operational and shared services** roles (Table 30).
- Global and larger firms allocate more resources to fundraising, shared services, and capital markets functions (Table 30).

8.3 Table 31: Use of external and internal advisers

What is measured. Whether the firm uses advisory boards, senior advisers/CEO councils, and strategy consultants.

Interpretation.

- A large fraction uses senior advisers/CEO councils and advisory boards (Table 31), supporting the view that PE builds operational expertise networks.
- Larger firms hire strategy consultants more frequently (Table 31), consistent with broader operational engineering investments.

Table 30

Private equity (PE) firm investment professionals.

This table reports the percentage of investment professionals in different specializations in the sample PE investor firms. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. HR=human resources.

Professionals by specialization	Mean	Median	AUM		IRR		Age		Offices	
			Low	High	Low	High	Old	Young	Local	Global
Deal professionals	54.2	50.0	57.7	51.2	55.1	59.2	59.3	50.4	58.7	48.6
Deal sourcing professionals	8.9	1.3	11.5	6.5	9.8	3.5**	6.0	11.1	9.8	7.7
Deal execution professionals	9.6	0.0	10.0	9.2	10.4	4.4	5.8	12.4**	11.5	7.2
Operating professionals	8.7	4.3	8.9	8.4	9.8	6.4	7.2	9.8	7.0	10.7
Consulting professionals	1.2	0.0	1.1	1.2	0.8	0.7	0.9	1.3	1.2	1.1
Shared services professionals	5.7	0.0	3.3	7.8	5.3	7.9	6.2	5.4	3.1	8.9**
Fundraising professionals	3.2	1.2	1.6	4.6***	4.0	2.6	2.9	3.4	1.7	5.1***
HR professionals for portfolio companies	0.4	0.0	0.4	0.3	0.2	0.3	0.2	0.5	0.2	0.6
Capital markets professionals	1.8	0.0	0.9	2.5	1.0	2.7	2.9	0.9	0.3	3.6**
Other	6.5	0.0	4.7	8.1	3.6	12.4	8.6	4.9	6.7	6.3
Number of responses	79	79	37	42	29	29	34	45	44	35

Table 31

Operational engineering assistance.

This table reports the percentage of the sample private equity (PE) investors who utilize a non-limited partner advisory board, senior advisors, or a chief executive officer (CEO) council or hire strategy consultants. The sample is divided into subgroups based on the median of assets under management (AUM), the internal rate of return (IRR) of most recent fund, the age of PE investor, and by whether PE investor has a global presence. Statistical significance of the difference between subgroup means at the 1%, 5%, and 10% levels are denoted by ***, **, and *, respectively. LP=limited partner.

	Mean	AUM		IRR		Age		Offices	
		Low	High	Low	High	Old	Young	Local	Global
Non-LP advisory board or group of advisors	38.0	32.4	42.9	27.6	48.3	38.2	37.8	47.7	25.7**
Senior advisors or CEO council or equivalent	48.1	35.1	59.5**	51.7	62.1	52.9	44.4	47.7	48.6
Hire strategy consultants to help with operating plans	31.6	18.9	42.9**	41.4	34.5	41.2	24.4	27.3	37.1
Number of responses	79	37	42	29	29	34	45	44	35

9 Heterogeneity: firm types, clusters, factors, and founder backgrounds (Tables 32–34)

9.1 Table 32: Cluster analysis (k-means)

What is measured. A $K = 3$ k-means partition on a set of strategy variables (leverage, capital structure logic, recruiting, proprietary sourcing, value drivers, operating partners, etc.).

Interpretation (as described by the paper).

- **Cluster 2** is most consistent with **operational and governance engineering**: higher reliance on trade-off logic, more CEO recruiting/replacement, more operating improvements and operating partner involvement (Table 32).
- **Cluster 1** is described as emphasizing **financial engineering** and “buy low/sell high,” with relatively low operational engineering; it also tends to contain smaller firms (Table 32).
- **Cluster 3** lies between the first two in operational vs financial emphasis (Table 32).

9.2 Table 33: Factor analysis

What is measured. Latent factors that explain co-movement of observed strategy variables.

How to read the loadings. A positive loading means the variable increases with the factor; a negative loading means it decreases.

Economic interpretation (combining the paper’s discussion and the reported loadings).

Table 32

Cluster analysis.

This table reports the results of a cluster analysis using partition clustering that divides “the observations into a distinct number of non-overlapping groups” (kmeans in Stata). The analysis generates three clusters. dtcap is debt to total capital; dteb is debt to EBITDA (earnings before interest, taxes, depreciation, and amortization); cstime is the sum of the rankings the private equity (PE) investors give to timing factors as important determinants of capital structure; cstrade is the sum of the rankings given to trade-off factors as important determinants of capital structure; invbus is the sum of the ranks given to the business model and the industry in an investment decision; invgmt is the rank given to the management team; invadv is the rank given to the ability to add value; recruitb is one if the PE investor typically recruits its own senior management team before investing; propdeal is the fraction of deals the PE investors claim are proprietary; growth equals one if the PE investor’s top ranking is growth in the value of the business; opimp equals one if the top ranking is operating improvements; costred is the fraction of deals the PE investor expects for cost reductions pre-investment; revgrow is the fraction of deals the PE investor expects for prospective revenue growth pre-investment; chceo is the fraction of deals the PE investor expects that changing the CEO is important pre-investment; buylowseilhi measures the importance of creating value buying low or selling high; multarb equals one if the general partner’s top ranking is industry-level multiple arbitrage; oppart is the fraction of deals that involve operating partners; and aum is assets under management.

Variable	Cluster #1		Cluster #2		Cluster #3		All	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
Dtcap	0.54	0.60	0.59	0.60	0.54	0.60	0.55	0.60
Dteb	3.64	3.50	3.95	4.00	4.11	4.00	3.88	4.00
Cstime	7.87	9.00	7.69	8.00	7.32	8.00	7.64	8.00
Cstrade	3.48	3.00	5.00	5.00	3.58	4.00	3.93	4.00
Invbus	7.83	8.00	7.38	7.50	8.47	8.00	7.91	8.00
Invgmt	4.04	4.00	3.75	3.50	3.79	4.00	3.88	4.00
Invadv	3.04	3.00	4.00	4.00	3.79	4.00	3.55	3.00
Recruitb	0.30	0.00	0.44	0.00	0.26	0.00	0.33	0.00
Propdeal	0.50	0.50	0.46	0.50	0.53	0.50	0.50	0.50
Growth	0.91	1.00	0.63	1.00	0.84	1.00	0.81	1.00
Opimp	0.04	0.00	0.25	0.00	0.16	0.00	0.14	0.00
Costred	25.17	20.00	55.38	55.50	31.42	20.00	35.55	25.00
Revgrow	73.04	80.00	82.19	87.50	66.16	70.00	73.31	80.00
Chceo	31.39	29.00	40.06	40.50	30.00	26.00	33.33	30.00
Buylowseilhi	181.48	175.00	190.38	190.00	68.74	70.00	147.00	150.00
Multarb	3.20	3.00	2.70	3.00	2.60	2.50	2.90	2.50
Opart	13.35	6.00	91.56	100.00	55.05	60.00	48.59	39.50
Aum	7.15	2.11	9.11	1.80	12.25	8.00	9.36	2.65
Number of firms	23	23	16	16	19	19	58	58

- **Factor 1: operational improvement orientation vs “good business” selection.** Positive loadings on leverage and operational variables (operating improvements, cost reductions, operating partners, “adding value”), and negative loadings on growth/proprietary business selection variables.
- **Factor 2: management replacement vs incumbent management.** Strong positive loadings on CEO change/recruiting and negative loadings on “investing in management” style variables.
- **Factor 3: financial engineering / scale.** Positive loadings on leverage, timing, and AUM, with negative loadings on operational improvement variables.

9.3 Table 34: Founder backgrounds and firm strategy

What is measured. How founder background clusters (finance vs operations vs prior PE) relate to strategy variables.

Interpretation.

- Firms founded by **finance-background** founders appear more aligned with financial engineering and investing with incumbent management.
- Firms with **prior PE** founders appear most strongly engaged in operational engineering (more focus on cost reduction, revenue growth, CEO change, and adding value).
- This supports a persistent “manager style” view: founder career histories shape the organization’s strategy (link to the broader “manager effects” idea in empirical corporate finance).

Table 33

Factor analysis.

This table reports the results of a factor analysis of sample private equity (PE) firm characteristics and answers. dtcap is debt to total capital; dteb is debt to EBITDA (earnings before interest, taxes, depreciation, and amortization); cstime is the sum of the rankings the PE investors give to timing factors as important determinants of capital structure; cstrade is the sum of the rankings given to trade-off factors as important determinants of capital structure; invbus is the sum of the ranks given to the business model and the industry in an investment decision; invmgt is the rank given to the management team; invaddv is the rank given to the ability to add value; recruitb is one if the PE investor typically recruits its own senior management team before investing; propdeal is the fraction of deals the PE investors claim are proprietary; growth equals one if the PE investor's top ranking is growth in the value of the business; opimp equals one if the top ranking is operating improvements; costred is the fraction of deals the PE investor expects for cost reductions pre-investment; revgrow is the fraction of deals the PE investor expects for prospective revenue growth pre-investment; chceo is the fraction of deals the PE investor expects that changing the CEO is important pre-investment; buylowseelli measures the importance of creating value buying low or selling high; multarb equals one if the general partner's top ranking is industry-level multiple arbitrage; oppart is the fraction of deals that involve operating partners; and aum is assets under management. Likelihood-ratio test: independent vs. saturated: $\chi^2(153) = 376.57$; $p\text{-value} = 0.0000$.

Panel A: Principal factors, unrotated (58 observations, 11 retained factors, 143 parameters)					
Factor	Eigenvalue	Difference	Proportion	Cumulative	
Factor #1	2.5695	0.6027	0.2659	0.2659	
Factor #2	1.9668	0.2159	0.2035	0.4695	
Factor #3	1.7509	0.4964	0.1812	0.6507	
Factor #4	1.2545	0.1910	0.1298	0.7805	
Factor #5	1.0635	0.3087	0.1101	0.8905	
Factor #6	0.7548	0.2473	0.0781	0.9687	

Panel B: Factor loadings						
Variable	Factor #1	Factor #2	Factor #3	Factor #4	Factor #5	Factor #6
Dtcap	0.5621	-0.0327	0.4938	0.0708	-0.2085	-0.2560
Dteb	0.5035	-0.0410	0.6676	-0.0047	-0.1817	0.0605
Cstime	0.1665	-0.2789	0.3737	0.2625	0.2806	-0.1203
Cstrade	0.1097	0.0423	-0.1426	0.1490	-0.0639	0.1176
Invbus	-0.2527	0.1049	0.0168	0.3620	-0.5344	-0.1266
Invmgmt	-0.2061	-0.2732	0.1520	-0.1401	0.5712	0.1638
Invaddv	0.5915	0.0930	-0.2394	-0.3619	-0.1367	0.1696
Recruitb	0.1495	0.5952	0.1109	0.2706	0.1373	0.0994
Propdeal	-0.3721	0.4450	-0.1385	0.2428	-0.0297	0.3292
Growth	-0.6203	0.3045	0.4265	-0.4185	-0.1212	-0.0479
Opimp	0.5183	-0.3698	-0.5246	0.3405	-0.0026	0.0244
Costred	0.5507	0.1235	0.0199	-0.3660	0.1529	-0.1378
Revgrow	-0.0509	0.5792	-0.1024	-0.1207	0.1880	-0.2658
Chceo	0.3892	0.6732	-0.2110	-0.0282	-0.0566	-0.0677
Buylowseelli	0.0983	0.3450	0.0160	0.2469	0.3541	-0.2450
Multarb	0.2566	0.2428	0.1569	0.3298	0.2231	0.1216
Oppart	0.3305	0.0754	-0.2566	-0.3633	-0.0318	0.1694
Aum	0.2185	0.1442	0.4313	0.0482	-0.0320	0.5167

Table 34

Relation of founder and firm characteristics.

This table reports the relation of founder characteristics to firm characteristics of the sample private equity (PE) investors. PE investors are classified as finance, operational, or other PE based on a cluster analysis on the fraction of their founders with financial, operational, or previous PE backgrounds. dtcap is debt to total capital; cstime is the sum of the rankings the PE investors give to timing factors as important determinants of capital structure; cstrade is the sum of the rankings given to trade-off factors as important determinants of capital structure; invbus is the sum of the ranks given to the business model and the industry in an investment decision; invmgt is the rank given to the management team; invaddv is the rank given to the ability to add value; recruitb is one if the PE investor typically recruits its own senior management team before investing; propdeal is the fraction of deals the PE investors claim are proprietary; growth equals one if the PE investor's top ranking is growth in the value of the business; opimp equals one if the top ranking is operating improvements; costred is the fraction of deals the PE investor expects for cost reductions pre-investment; revgrow is the fraction of deals the PE investor expects for prospective revenue growth pre-investment; chceo is the fraction of deals the PE investor expects that changing the chief executive officer is important pre-investment; buylowseelli measures the importance of creating value buying low or selling high; and aum is assets under management.

Variable	Mean (all)	Finance	Operations	Other PE
Dtcap	0.56	0.52	0.58	0.57
Cstime	7.47	7.33	7.76	7.00
Cstrade	3.86	3.26	4.64	3.22
Invbus	7.77	7.44	8.20	8.56
Invmgmt	3.82	4.26	3.56	2.89
Invaddv	3.63	3.37	3.52	4.22
Recruitb	0.31	0.30	0.32	0.22
Propdeal	0.51	0.54	0.46	0.47
Growth	0.80	0.78	0.80	0.78
Opimp	0.13	0.11	0.12	0.22
Costred	35.6	25.0	38.1	42.3
Revgrow	70.3	66.6	70.5	80.6
Chceo	30.6	26.3	30.4	48.8
Buylowseelli	140.9	155.4	150.2	125.0
Aum	9.55	10.30	11.06	4.72
Year_founded	1993.5	1992.7	1994.5	1994.2
Number of responses	79	27	25	9

10 Concluding synthesis

1. **PE valuation practice is hurdle-based.** IRR/MOIC dominate, DCF/CAPM are rarely used directly (Table 5, Table 8).
2. **The terminal value is central.** Standard five-year horizons and heavy reliance on multiples-based terminal values imply that exit assumptions drive valuation (Fig. 1, Table 7).
3. **Capital structure reflects both fundamentals and timing.** Interest rates and industry conditions are top-ranked; tax trade-off and “max debt” motives are secondary but present (Tables 12–13).
4. **Governance engineering is real and frequent.** PE typically controls boards and actively advises; CEO replacement is material (Tables 14–18).
5. **Operational value creation is emphasized.** Growth and operational improvements rank above leverage engineering; post-deal actions show increased cost reduction and management change relative to pre-deal expectations (Tables 22–26).
6. **PE strategies are heterogeneous and path-dependent.** Clusters and factors reveal distinct styles; founder backgrounds correlate with these styles (Tables 32–34).

11 Conclusion

11.1 Summary

This paper concludes that private equity (PE) firms describe their investing process in a way that is both **systematic** and **meaningfully different from standard textbook corporate-finance decision rules**. Using a survey of PE investors, the authors document what PE firms *say they do* in valuation, capital-structure choice, governance, and operational value creation, and they show that there is substantial cross-firm heterogeneity in these practices.

The core conclusions are:

- (i) **Deal evaluation is dominated by IRR/MOIC and multiples; DCF/NPV is rarely used.** PE investors report relying primarily on internal rate of return (IRR) and multiples of invested capital (MOIC), with terminal value typically implemented via exit multiples rather than a full discounted-cash-flow (DCF) apparatus. Very few report using NPV/DCF techniques as a primary valuation tool.
- (ii) **PE hurdle rates are not CAPM/WACC-based and vary across firms.** Most PE firms do not use CAPM-style cost-of-capital methods to set discount rates. Instead, they target investment returns well above CAPM-implied rates and adjust their target IRRs using firm-specific, deal-specific, and macro considerations. This implies that different PE firms can apply different effective hurdle rates to the same deal opportunity.

- (iii) **Limited partners (LPs) are perceived to benchmark PE using absolute performance.** Respondents report that LPs care more about absolute performance measures (e.g., net IRR, cash-on-cash) than about public-market-relative “alpha” style benchmarks. The authors note that wider adoption of risk-adjusted PE benchmarking tools (e.g., PME-style metrics) could shift this perception over time.
- (iv) **Capital-structure choice mixes trade-off and market-timing logic.** PE investors report that portfolio-company leverage decisions reflect both (i) factors aligned with classic trade-off views (e.g., debt capacity, firm fundamentals) and (ii) market-timing considerations (e.g., debt-market conditions). Hence the survey points to a hybrid decision process rather than a single canonical theory.
- (v) **PE firms describe themselves as active owners emphasizing governance and operations.** PE investors report strong reliance on incentive alignment (meaningful management equity), active board design (small boards with PE participation), and hands-on involvement. They emphasize value creation through revenue growth, governance/incentives, improving exit value, add-on acquisitions, management replacement when needed, and cost reductions. Many firms invest in internal operating resources and external advisers to implement operational changes.
- (vi) **There is large heterogeneity in PE “styles,” correlated with founder backgrounds.** Exploratory cluster and factor analyses suggest that financial engineering, governance engineering, and operational engineering co-vary in different ways across firms: some emphasize operational engineering, others rely more on management replacement, etc. These strategic differences correlate with founders’ career histories, suggesting persistent organizational “style.”

11.2 Why PE practice differs from textbook finance

The paper’s findings are naturally interpreted through frictions and incentives that are specific to private equity:

1) IRR/MOIC are “contract-consistent” decision tools. PE is organized around finite-life funds, delegated management, and performance reporting that is itself IRR- and multiple-centric. Even if DCF/NPV is theoretically appealing, IRR/MOIC can be a robust and communicable decision rule that aligns with how deals are marketed, monitored, and evaluated inside the GP–LP relationship.

2) Precise discount-rate estimation is unusually difficult in PE. DCF requires (i) credible long-run cash-flow forecasts and (ii) a well-defined risk-adjusted discount rate. In PE, illiquidity, leverage, nonlinear payoff structures, and severe asymmetric information can make WACC/CAPM-based discounting both hard to implement and potentially misleading in practice. This pushes practitioners toward heuristics (multiples) and hurdle rates rather than formal discount-rate estimation.

3) “Engineering” is partly an organizational capability choice. If value creation depends on governance and operating interventions, then a PE firm’s human capital (operating partners, advisor networks, sourcing capabilities, management-replacement ability) becomes a key asset. Different founder backgrounds and organizational histories plausibly generate persistent differences in these capabilities, which rationalizes the observed heterogeneity in PE strategies.

4) Absolute benchmarking is plausible given LP objectives. Many LPs (e.g., pensions) have obligations framed in absolute terms, which can rationalize a focus on absolute PE performance. But as risk-based benchmarking becomes more widespread and standardized, PE performance evaluation could shift toward more systematic relative measures.

11.3 Takeaway

Survey evidence suggests PE firms primarily use IRR/MOIC and multiples rather than DCF/CAPM, choose leverage using a mix of trade-off and market-timing logic, emphasize active governance and operational value creation, and exhibit strong cross-firm heterogeneity in strategy that is linked to founder backgrounds.